Trung Dang

765 491 6622
dddtrung@cs.utexas.edu
kuroni.github.io



Education

08/2023 – Now **Ph.D., Computer Science**, *The University of Texas at Austin*, Austin, TX, Advisor: Shuchi Chawla
Research Interest: Theoretical Computer Science, with a focus on Algorithm Design under Data Uncertainty.

08/2019 – 05/2023 **B.Sc., Computer Science**, *Purdue University*, West Lafayette, IN
Coursework: Randomized Algorithm. Algorithmic Economics. Intro to Artificial Intelligence. Machine Learning.

Publications and Manuscripts

Publications

[1] Shuchi Chawla, Dimitris Christou, **Trung Dang**, and Zhiyi Huang. Pricing pandora's boxes: Revenue maximization in sequential information acquisition. In *18th Innovations in Theoretical Computer Science Conference (ITCS 2027)*, 2027.

[2] Kamyar Azizzadenesheli, **Trung Dang**, Aranyak Mehta, Alexandros Psomas, and Qian Zhang. Reward selection with noisy observations. In *International Conference on Algorithmic Learning Theory*. PMLR, 2026.

[3] Shuchi Chawla, Dimitris Christou, **Trung Dang**, Zhiyi Huang, Gregory Kehne, and Rojin Rezvan. A multi-dimensional online contention resolution scheme for revenue maximization. In *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 1720–1757. SIAM, 2025.

[4] **Trung Dang** and Zhiyi Huang. Robust max selection. In *2025 IEEE International Symposium on Information Theory (ISIT)*, pages 1–6. IEEE, 2025.

[5] Shuchi Chawla, **Trung Dang**, Zhiyi Huang, and Yifan Wang. Multi-unit combinatorial prophet inequalities. In *International Conference on Web and Internet Economics*, 2025.

[6] Shuchi Chawla, Dimitris Christou, and **Trung Dang**. Commitment gap via correlation gap. In *International Conference on Web and Internet Economics*, 2025.

[7] **Trung Dang**, Jasper Lee, Maoyuan Song, and Paul Valiant. Optimality in mean estimation: Beyond worst-case, beyond sub-gaussian, and beyond $1+\alpha$ moments. In *Advances in Neural Information Processing Systems*, volume 36, pages 4150–4176, 2023.

[8] Zitao Li, **Trung Dang**, Tianhao Wang, and Ninghui Li. Mgd: A utility metric for private data publication. In *Proceedings of the 8th International Conference on Networking, Systems and Security*, NSysS '21, pages 106–119, New York, NY, USA, 2021. Association for Computing Machinery.

Manuscripts

[9] Gagan Aggarwal, **Trung Dang**, Aviad Rubinstein, and Mingfei Zhao. The no-menu complexity of revenue-maximizing auctions. *In Submission to STOC 2027 - 59th ACM Symposium on Theory of Computing*, 2026.

[10] Shuchi Chawla and **Trung Dang**. Single-sample prophet inequalities: A combinatorial to single-item reduction. *In Submission to STOC 2027 - 59th ACM Symposium on Theory of Computing*, 2026.

[11] **Trung Dang**, Daniel Halpern, Anuran Makur, Alexandros Psomas, Japneet Singh, and Paritosh Verma. Fair division under inaccurate preferences. *arXiv preprint arXiv:2602.24169*, 2026.

[12] **Trung Dang**, Walter McKelvie, Paul Valiant, and Hongao Wang. Improving pearson's chi-squared test: hypothesis testing of distributions–optimally. *arXiv preprint arXiv:2310.09408*, 2023.

Industry Experience

06/2026 – 09/2026 **Student Researcher**, *Google*, Mountain View, CA
- Performed theoretical research on market design, mechanism design, and efficient algorithm design.

05/2022 – 08/2022 **Software Engineer Intern**, *Meta*, Seattle, WA
- Performed migration of an important internal data streaming service to a new framework.
- Implemented additional features to integrate the streaming service more smoothly into the new framework.

Scholarships & Awards

2022 **Honorable Mention** for the CRA Outstanding Undergraduate Researcher Awards 2023.

2025 **Receiver** of the Calhoun Fellowship.

2022 20th place at ICPC World Finals 2021.

2024 31st place at ICPC World Finals 2023.

2021 **First Prize Winner** of the NIST "A Better Meter Stick for Differential Privacy" competition.

Teaching

Fall'26 **C S 388D: Data-Driven Algorithm Design**, *Teaching Assistant*, UT Austin

Fall'24 – Spring'26 **C S 331(H): Algorithms and Complexity (Honors)**, *Teaching Assistant*, UT Austin

Fall'23 – Spring'24 **C S 375: Compilers**, *Teaching Assistant*, UT Austin